

مختبر السيركت / م. صفاء مقبل

نمط الامتحانات ::

((الميد)) :

الميد عبارة عن 30 علامة [10 عملي + 20 نظري (حسابي)]

الاسئلة الحسابية عبارة عن دوائر بسيطة تحل عن طريق قانون أوم ، ميش ، نود.. الخ
(نموذج امتحان ميد) :

امتحانات لاب سيركت

السؤال الأول ← توصيل الدارة التالية عملياً وإيجاد المقاومة المكافئة والتيارات والجهود فيها (١٠ علامات)

السؤال الثاني ← جد المقاومة المكافئة والتيارات والجهود والقوة المستهلكة في المقاومة (2.2k) حسابياً في الدارة السابقة.

السؤال الثالث ← جد R_{th} ، V_{th} بين الطرفين a, b

السؤال الرابع ← حول توصيل star التالي Delta

((التقارير)) :

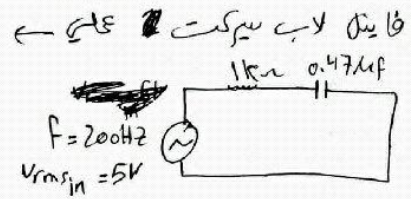
عليهم 20 علامة

((الفاينل)) :

الامتحان [عملي 25 علامة + نظري 25 علامة]

* نموذج عملي:

قم بتوصيل الدارة التالية واحسب ما يلي :



Q1 →

find →

V_{p-p} ?

V_p ?

I_{rms} ?

$V_{R,rms}$?

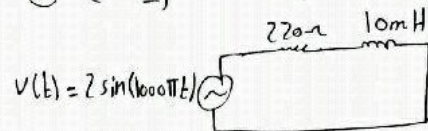
~~φ~~

T ?

Q2 → find the phase shift between V_c and V_{in}

* نظري :

فايند لاب سيركيت ← نظري ①



Q2 →

find →

1) f 2) T 3) X_L

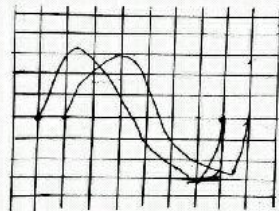
4) Z 5) I 6) V_R

7) V_L 8) θ 9) p.f 10) V_{rms}

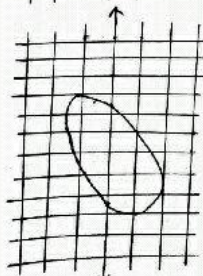
Q2 → using equation explain the relation between →

1) I, f 2) p.f, f

Q3 → Find the phase shift for the following →

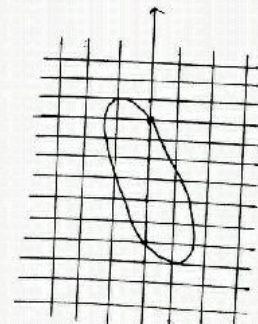


$$\theta = \frac{D_2}{D_1} \times 360 = \frac{1}{8} \times 360 = 45^\circ$$



$$\theta = 180 - \sin^{-1}\left(\frac{5}{8}\right)$$

$$\therefore \theta = 123.56^\circ$$



$$\theta = \sin^{-1}\left(\frac{6}{8}\right)$$

$$\therefore \theta = 48.59^\circ$$

قائىل لاب سىگنل نظري ← (2)

Q 4 → Vertical sensitivity = 50mV/division
Horizontal sensitivity = 0.2μs/division

Find →

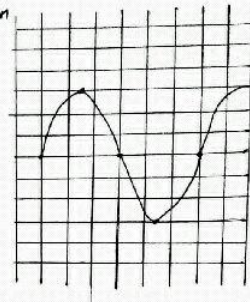
1) Peak value

$$3 \times 50m = 150mV$$

2) Frequency (f)

$$T = 6 \times 0.2\mu = 1.2\mu s$$

$$f = \frac{1}{T} = 833333.33 \text{ Hz}$$



3) Reading voltmeter

$$V_{rms} = \frac{V_m}{\sqrt{2}} = \frac{150m}{\sqrt{2}} = 0.106 \text{ V}$$

4) Find the number of division when $V_{p-p} = 200mV$

$$200m = y \times 50m \rightarrow \therefore y = 4$$

5) Find the number of division when $T = 10\mu s$

$$10\mu = y \times 0.2\mu \rightarrow \therefore y = 50$$

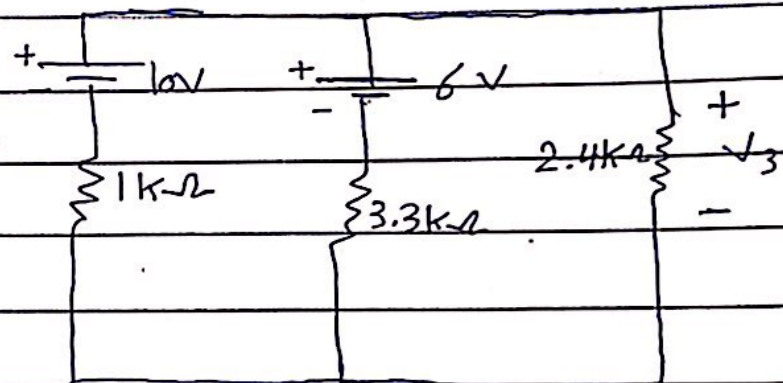
Q 5 → when $f = \infty$ in RC circuit, find →

1) V_c

2) P.f

Q) Consider the circuit given below:-

- using mesh current to determine the Voltage V_3 .
- Determine the Thevenin's equivalent circuit seen from the terminals of R_3 (V_2).



Sol:-

a. $1kI_1 - 10 + 6 + 3.3kI_1 - 3.3kI_2 = 0$

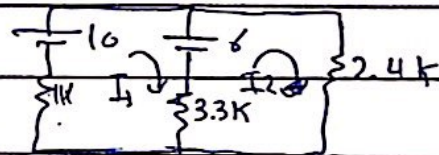
$4.3kI_1 - 3.3kI_2 = 4$... ①

$3.3kI_2 - 3.3kI_1 + 6 + 2.4kI_2 = 0$

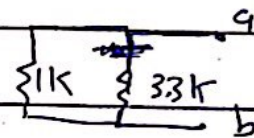
$-3.3kI_1 + 5.7kI_2 = 6$... ②

$I_1 = 3.12 \text{ mA}$

$I_2 = 2.86 \text{ mA}$



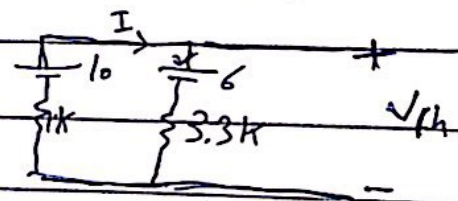
b. $R_{th} = \frac{3.3 \times 1}{3.3 + 1} = 767 \Omega$



$V_{th} \Rightarrow 4.3I + 6 - 10 = 0$

$I = 0.93 \text{ mA}$

$V_{th} = 6 + 3.3 \times 0.93 = 9.1 \text{ V}$



Don't miss the change..
Be part of it !!

1

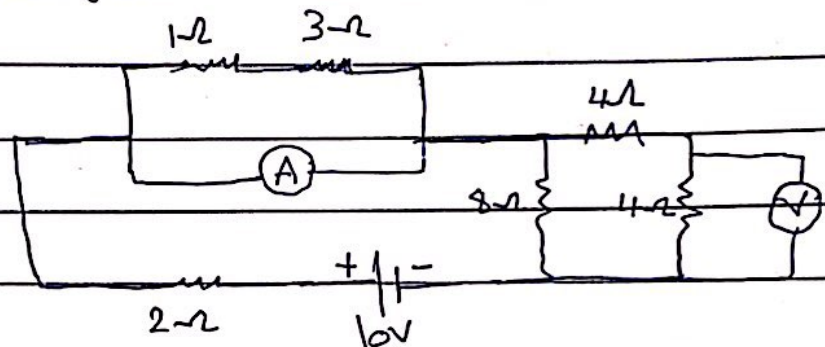


Q) In the figure below find :-

a. Equivalent Resistance.

b. Voltmeter Reading.

c. Ammeter Reading.



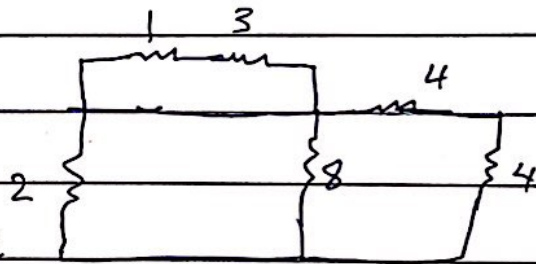
Sol :-

a. (4//4) Series

$$4 + 4 = 8$$

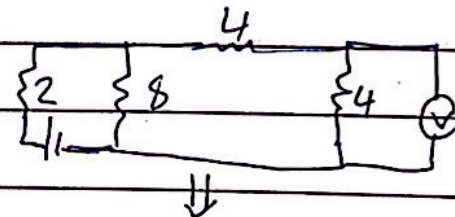
$$(8//8) \Rightarrow \frac{8 \times 8}{8 + 8} = 4\Omega$$

$$(4, 2) \text{ series} \rightarrow 4 + 2 = 6\Omega$$



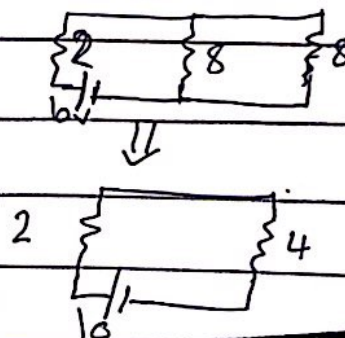
b. $V_{4\Omega} = \frac{4 \times 10}{2 + 4} = 6.66V$

The Voltmeter reading = $3.33V$



c. Ammeter Reading :

$$I = \frac{10}{6} = 1.67A$$

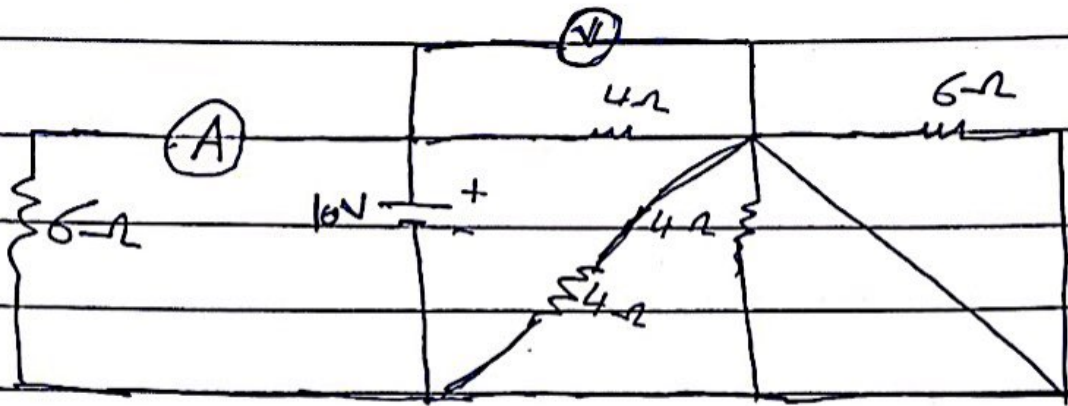


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2

PEE

Q) Find R_{eq} , Reading Voltmeter and Reading Ammeter.



Sol:-

NP is $4, 4, 6$

• $R_{eq} = \frac{4 \times 6}{4 + 6} = 2.4 \Omega$

• Voltmeter Reading = 10 Volt

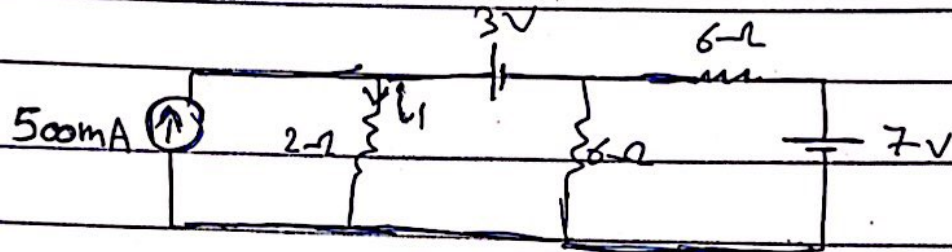
• Ammeter reading = $\frac{10}{6} = 1.67 A$

3

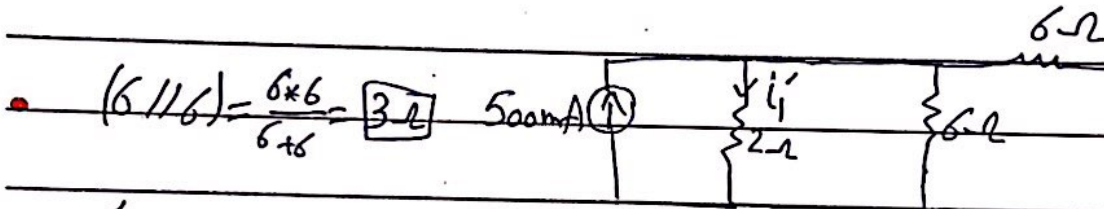


Don't miss the change..
Be part of it !!

Q) Determine i_1 using superposition method.

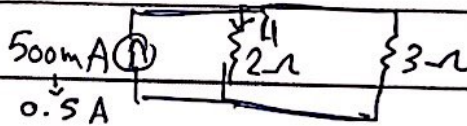


Sol :-



$$I_1' = \frac{3 \times 0.5}{3 + 2}$$

$$I_1' = 0.3 \text{ A}$$

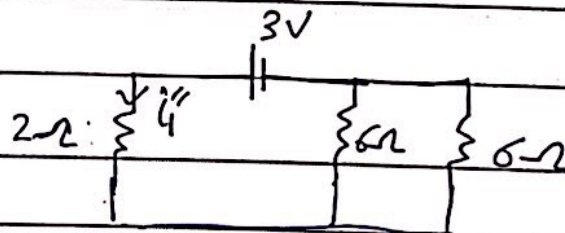


$$6 \parallel 6 = \frac{6 \times 6}{6 + 6} = 3 \Omega$$

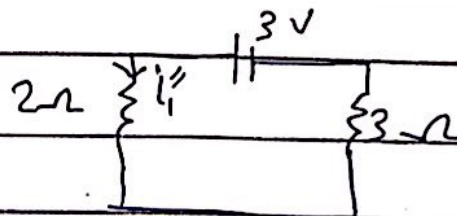
$$I_1'' = \frac{V}{R}$$

$$I_1'' = \frac{3}{5}$$

$$I_1'' = 0.6 \text{ A}$$



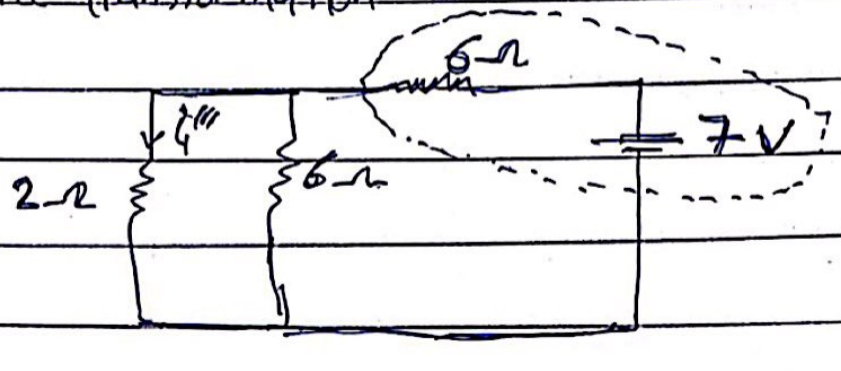
$$6 \parallel 6 = 3$$



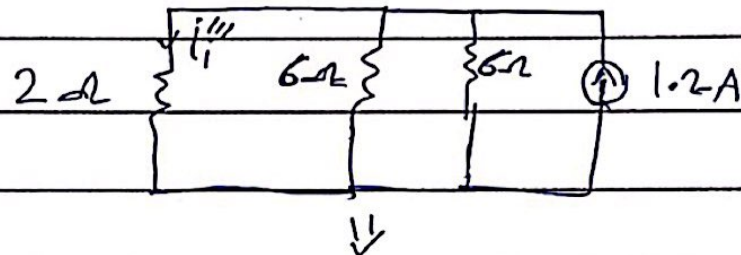
• Using source transformation

$$I = \frac{V}{R} = \frac{7}{6}$$

$$I = \frac{7}{6} \approx 1.2 \text{ A}$$

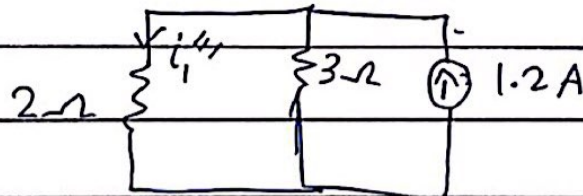


$$6 \parallel 6 = 3$$



$$I_1''' = \frac{1.2 \times 3}{2 + 3}$$

$$I_1''' = 0.72 \text{ A}$$



$$I_1 = I_1' + I_1'' + I_1'''$$

$$I_1 = 0.3 + 0.6 + 0.72$$

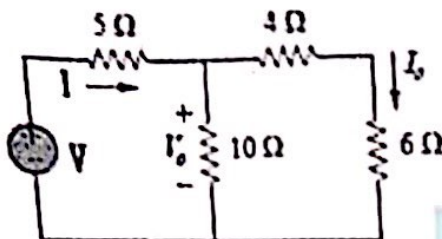
$$I_1 = 1.62 \text{ A}$$



Student Name:

For each question, choose the correct answer

Consider the circuit shown below:



-The equivalent resistance seen by the source (R_{eq}) is:

- a. 35Ω b. 15Ω c. 8.75Ω d. 10Ω

-The value of the voltage V_o , when $V = 18V$, is

- a. $6V$ b. $12V$ c. $9V$ d. $13.5V$

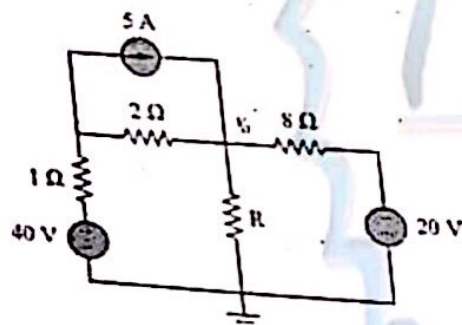
-The value of the current I_o , when $I = 0.9A$, is

- a. $0.45A$ b. $0.6A$ c. $0.3A$ d. $0.675A$

-When the current $I_o = 0.4A$, then the value of I is

- a. $0.533A$ b. $0.6A$ c. $1.2A$ d. $0.8A$

Consider the circuit shown below, where $R = 4\Omega$.



-The current through the 1Ω resistor is:

- a. $10A$ b. $11.74A$ c. $9.11A$ d. $7.95A$

-The current through the 8Ω resistor is:

- a. $5.33A$ b. $5.77A$ c. $4.34A$ d. $5A$

-The power absorbed by resistor R is

- a. $85.64W$ b. $100W$ c. $57.03W$ d. $109.22W$

-In Superposition theorem, while considering a source, all other voltage sources are:

- a. open circuited b. short circuited c. change its position d. removed from the circuit

-In Superposition theorem, while considering a source, all other current sources are:

- a. open circuited b. short circuited c. change its position d. removed from the circuit

-Which of the following series combinations dissipates the most power when connected to a source:

- a. One 100Ω resistor b. Two 100Ω resistors c. Three 100Ω resistors d. Four 100Ω resistors

-If one of the resistors in a parallel circuit is removed, the total resistance:

- a. Decreases by the value of the removed resistor
b. Remains the same
c. Increases
d. Non of the above

-The Thevenin equivalent voltage for a given circuit is found by:

- a. Shorting the output terminals
b. Opening the output terminals
c. Shorting the voltage source
d. Opening the voltage source

-When measuring current drawn from a DC power supply, it is true to say that the meter will act in circuit as:

- a. A perfect conductor b. An open resistance c. An insulator d. An extra current drain

-What does a multimeter measure?

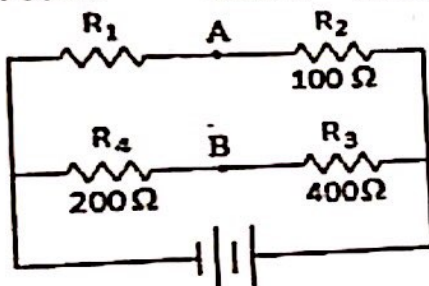
- a. Resistance, capacitance and inductance
b. Voltage, current and resistance
c. Resistance and reactance
d. SWR and power

-In measuring volts and amperes, the connections should be:

- a. The voltmeter in series and ammeter in parallel
b. The voltmeter in parallel and ammeter in series
c. Both voltmeter and ammeter in parallel
d. Both voltmeter and ammeter in series

-For the circuit shown below, $V_{AB}=0V$, the value of R_1 :

- a. 100Ω b. 50Ω c. 200Ω d. 800Ω



-When a sine wave has a frequency of 60Hz. In 10sec it goes through:

- a. 600 cycles b. 60 cycles c. 6 cycles d. 6000 cycles

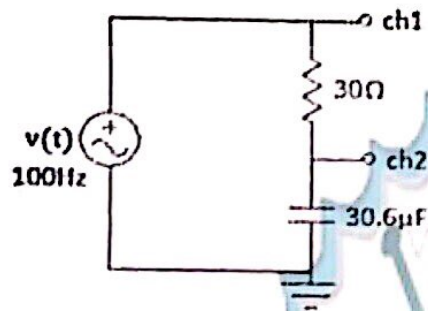
-A sinusoidal voltage is applied across an inductor, when the frequency of the voltage is increased, the current:

- a. Decreases b. Increases c. Remains constant d. Non of the above

-A capacitor and a resistor are connected in series to a sine wave generator. The frequency is set so that the capacitive reactance is equal to the resistance and thus, an equal amount of voltage appears across each component. If the frequency is decreased:

- a. $V_R > V_C$ b. $V_C > V_R$ c. $V_R = V_C$

Consider the circuit shown below:



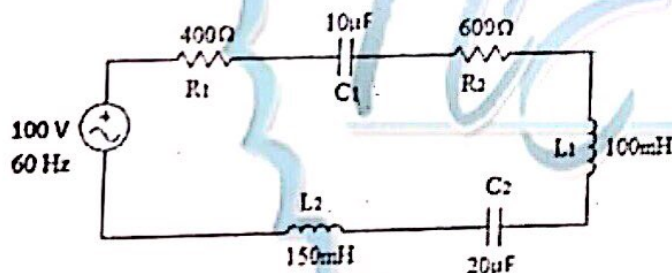
-The measured angle from both ch1 and ch2 is between:

- a. V_i and V_c b. V_i and V_R c. I_i and V_c d. I_i and I_c

-The phase shift for the circuit shown below:

- a. 60° b. 30° c. 45° d. 90°

Consider the circuit shown below:



-The amplitude of the impedance is:

- a. 1045Ω b. 1000Ω c. 1095Ω d. 100Ω

-The phase shift angle for this load:

- a. 22.6° b. -22.6° c. -16.8° d. 16.8°

-The power factor is:

- a. 0.95 lagging b. 0.95 leading c. 0.92 lagging d. 0.92 leading

-Total current is:

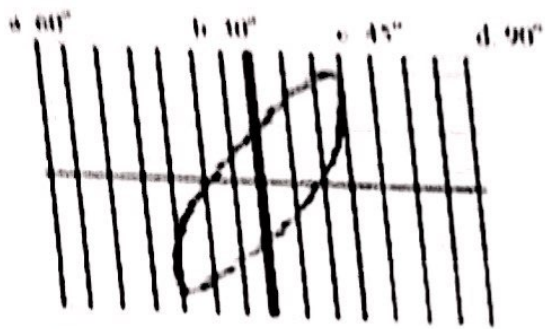
- a. 75mA b. 55mA c. 95mA d. 115mA

-Voltage across L_2 is:

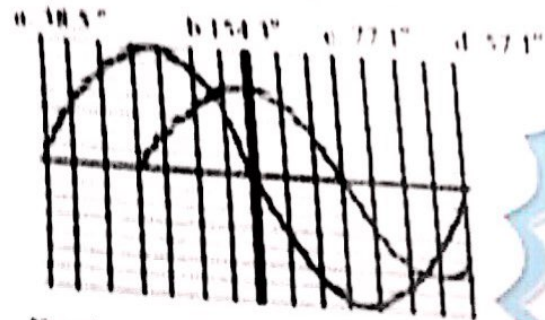
- a. 5.3v b. 4.2v c. 3.1v d. 6.4v

3

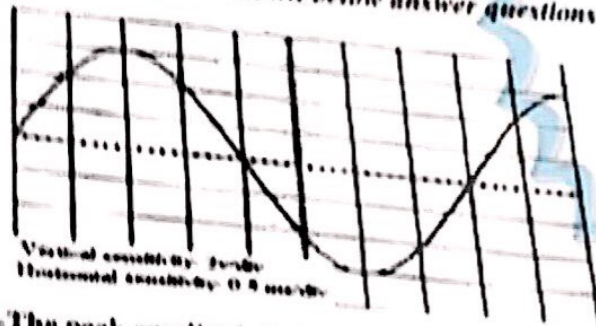
The phase shift for the figure below



The phase shift for the figure below

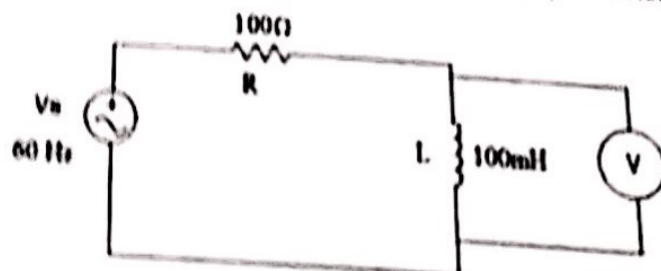


For the waveform shown below answer questions



- The peak amplitude is (.....)
- The time period is (.....)
- The frequency is (.....)
- The RMS value is (.....)

For the circuit shown below the reading of voltmeter is 5V



- The voltage across resistance is:
 - a. 5V b. 13.27V c. 14.18V d. 18.27V
- The source voltage is:
 - a. 5V b. 13.27V c. 14.18V d. 18.27V